

Smart Drone With Real-Time Face Recognition

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What if you could make a drone that would recognise people, track someone in a crowd, and help find criminals? What if this drone could also detect traffic congestion on the road ahead, or detect people stuck in floods, earthquakes, or similar disaster situations? This smart AI-enabled drone can do all these tasks. It can shoot video in real time, process the video, recognise people, and detect traffic and crowd during public rallies and gatherings.

This is a low-cost quadcopter drone using an F450 frame and 1000kv brushless motors. It is equipped with a Raspberry Pi 4 (Rpi 4) board to process images independently without affecting the flight controller.

The author's prototype is shown

Components Required

Table 1

Components	Quantity	Description
Quadcopter frame	1	F450
DC motor	4	1000kv brushless
Flight controller	1	KK2.2
Electronic speed control (ESC)	4	30A
Radio transmitter and receiver	1	FlySky FS-CT6B
LiPo battery	1	3S 11.2V, 2200mAh
AA battery	8	1.5V alkaline battery
Propeller (CW)	2	25.4cm (10-inch) 1045 clockwise propeller
Propeller (CCW)	2	25.4cm (10-inch) 1045 counter-clockwise propeller
XT60	1	Connector for LiPo battery
Raspberry Pi 4 (Rpi 4)	1	Single-board computer
Camera	1	Raspberry Pi 5MP CSI
Audio amplifier	1	PAM8403
Speakers	2	Portable



Fig. 1: Author's prototype of the quadcopter drone



Fig. 3: Raspberry Pi 4 board mounted on the drone frame



Fig. 2: Live image being captured by camera mounted on the drone



Fig. 4: RPi camera mounted on the drone frame